

Lab Manual: Deploying a Streamlit App to Google App Engine with Docker

Objective

By the end of this lab, you will:

- Package a Streamlit app into a Docker container.
- Configure Google App Engine with a custom runtime.
- Deploy the app on GCP and access it via a public URL.

Prerequisites

- A working **Streamlit app** (`app.py`).
- Python 3.7+ installed locally.
- **Docker** installed and running.
- **Google Cloud SDK (gcloud CLI)** installed and authenticated (`gcloud init`).
- A GCP project created and set as active.

Step 1: Prepare Your Streamlit App

1. Ensure your main app file is named: `app.py`
2. Create a `requirements.txt`: `pip freeze > requirements.txt`

This captures all Python dependencies for the container.

Step 2: Create a Dockerfile

Inside your project folder, create a file named Dockerfile:

```
# Use lightweight Python image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy all app code
COPY . .

# Expose Streamlit port
EXPOSE 8080

# Start Streamlit app on port 8080
CMD ["streamlit", "run", "app.py", "--server.port=8080", "--server.address=0.0.0.0"]
```

Explanation:

- `python:3.9-slim` → lightweight base image.
- `WORKDIR /app` → sets working directory.
- `pip install --no-cache-dir` → installs dependencies efficiently.
- `EXPOSE 8080` → App Engine requires port 8080.
- `CMD [...]` → runs Streamlit bound to `0.0.0.0:8080` so it's accessible externally.

Step 3: Create App Engine Configuration

Create a file named `app.yaml`:

```
runtime: custom
env: flex
```

Explanation:

- `runtime: custom` → tells App Engine to use your Dockerfile.
- `env: flex` → deploys in App Engine's **Flexible Environment**.

Step 4: Deploy to Google App Engine

Run the following command in your project directory:

```
gcloud app deploy
```

- Builds the Docker image.
- Pushes it to **Google Container Registry (GCR)**.
- Deploys it to **App Engine**.

Step 5: Open the Deployed App

Once deployment is complete, run: `gcloud app browse`

This opens your app in the browser at a URL like:

<https://your-project-id.uc.r.appspot.com>

Verification Checklist

- `app.py` exists.
- `requirements.txt` generated.

- Dockerfile created with correct CMD.
- app.yaml created with custom runtime.
- Deployment runs successfully with `gcloud app deploy`.
- App accessible via public App Engine URL.

Summary

In this lab, you:

- Dockerized your **Streamlit app**.
- Configured **App Engine flexible environment**.
- Deployed using `gcloud app deploy`.
- Accessed your app via a public URL.

This approach gives you **more control** than Streamlit Cloud, you can install system packages, run background jobs, and scale with GCP infrastructure.